

THIRD SEMESTER**B.Tech. (ECE)**

END SEM. EXAMINATION, Nov.-Dec. 2025

Course Code: ECECC305/EIECC305/CECC305/VTECC305

Course Title: Digital Circuits and Systems

Time: 03:00 Hrs.

Max. Marks: 50

Note: Attempt all the five questions. All questions carry equal marks. Missing data/information if any will be suitably assumed and mentioned in the answer.

Q. No.	Question	Marks	CO															
Q1	Attempt any 2 parts of the following																	
1(a)	(i) Obtain the minimized expression for a combinational circuit for BCD to 2421 code conversion. (ii) Implement it using NOR gates only	(4+1)	CO1															
1(b)	(i) Write the truth table for Gray-to-BCD code conversion (ii) Design the corresponding circuit using 4x1 multiplexers and basic logic gates.	(3+2)	CO1															
1(c)	Design a combinational logic circuit that detects the presence of any of the six illegal code groups in the 84(-2)(-1) code (used for the binary representation of digits 0 to 9) and provides a logic '1' output when any illegal code is detected. Implement the circuit using a decoder.	(5)	CO1															
Q2	Attempt any 2 parts of the following																	
2(a)	Solve the following expression using K-map: $F(A,B,C,D,E) = \sum m(1,3,4,7,8,9,12,15,17,19,23,24,25,28)$; $d(A,B,C,D,E) = \sum m(0,2,6,10,13,29,30)$ i) SOP form ii) POS form	(2.5+2.5)	CO2															
2(b)	Design a combinational circuit to compare the magnitudes of two 2-bit numbers. Implement the designed magnitude comparator using a PROM of suitable size.	(5)	CO2															
2(c)	Design a BCD Adder using 4-bit parallel adders. Explain its working with necessary logic diagrams.	(5)	CO2															
Q3	Attempt any 2 parts of the following																	
3(a)	Using positive edge-triggered JK flip-flops, design a synchronous Mod-5 counter whose counting sequence is as follows: $110 \rightarrow 101 \rightarrow 001 \rightarrow 111 \rightarrow 011 \rightarrow 110$	(5)	CO3															
3(b)	Design a 4-bit register, incorporating 4 multiplexers and 4 positive edge-triggered D flip-flops, having the behavior specified by the table given below: <table><tr><th>Select Line S0</th><th>Select Line S1</th><th>Register Operation</th></tr><tr><td>0</td><td>0</td><td>Circular shift right</td></tr><tr><td>0</td><td>1</td><td>Hold</td></tr><tr><td>1</td><td>0</td><td>Synchronous reset</td></tr><tr><td>1</td><td>1</td><td>Complement content</td></tr></table>	Select Line S0	Select Line S1	Register Operation	0	0	Circular shift right	0	1	Hold	1	0	Synchronous reset	1	1	Complement content	(5)	CO3
Select Line S0	Select Line S1	Register Operation																
0	0	Circular shift right																
0	1	Hold																
1	0	Synchronous reset																
1	1	Complement content																
3(c)	Using negative edge triggered S-R flip flop, design a sequential circuit for the State Diagram given below:	(5)	CO3															

Q4	Attempt any 2 parts of the following		
4(a)	Draw the gate-level implementation of 3 X 8 decoder using basic gates and write the Verilog code for its implementation using structural modeling.	(5)	CO4
4(b)	Write a Verilog module for a 3-bit asynchronous up-counter with asynchronous reset using behavioral modeling. Do mention the possible comments explaining the logic used alongside the code for better understanding of the implementation.	(5)	CO4
4(c)	Draw the state diagram and circuit of 3-bit Johnson counter.	(5)	CO4
Q5	Also write the Verilog code for the counter.		
5(a)	Attempt any 2 parts of the following		
	Explain the working of a R-2R ladder DAC with circuit diagram and derive the output voltage expression for the circuit.	(5)	CO5
5(b)	(i) Draw the MOSFET-based circuit implementing the Boolean expression: $Y = A + B + C$ using the CMOS logic circuit family, and explain its functioning. (ii) Explain the following terms related to logic families: Propagation Delay, Noise Margin, Power Dissipation, and Fan-out.	(3+2)	CO5
5(c)	Implement the circuit defined by following equations: $F_1 = \sum(0,1,3,5)$ and $F_2 = \sum(3,5,7)$ using a PLA with 3 inputs and minimum product terms. Draw the PLA table and the circuit diagram.	(5)	CO5